On the Privacy Risks of Model Explanations

Seminar report on Shokri, Strobel and Zick, *Proc. AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2021

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Summary. Model explanations are promoted as a way to make machine-learning decisions transparent, yet every explanation is computed from a model that was trained on private data. Shokri, Strobel and Zick [1] ask whether an adversary can exploit explanations to infer which records were in the training set. Using membership inference attacks that threshold the variance of an explanation, they show that gradient-based explanations leak membership on high-dimensional tabular data (up to 67% attack accuracy against a 50% baseline), that perturbation-based explanations such as SmoothGrad and LIME do not leak in their experiments, and that example-based explanations leak severely: a training point often explains itself, nearly the entire training set can be reconstructed in one case, and minority groups are disproportionately exposed. Below I go through the method and results, discuss where I think the evidence is weaker than the headline numbers suggest, and relate the paper to Bertran et al. [16], the paper I was assigned to ask questions about, and answer the questions I was asked in the seminar.

1 Introduction

Machine-learning models are increasingly used for decisions about credit, health care and employment. Their complexity makes such decisions hard to contest, which has led to calls for automated explanations, most prominently in the debate about a "right to explanation" under the EU General Data Protection Regulation [17]. Google, Microsoft and IBM all offer explanation toolkits with their machine-learning platforms, so explanations are already something end users get to see.

From a data-privacy point of view, anything released about a model trained on personal data can leak information about that data. The standard way to measure such leakage is membership inference [2]: an adversary tries to decide whether a specific record was part of the training set. Membership alone can be sensitive (being in the training set of a model for a particular disease reveals the diagnosis), and the UK Information Commissioner's Office lists membership inference as a threat in its AI auditing guidance. Before this paper, the privacy cost of explanations had barely been studied; the closest work showed that gradient explanations make it easier to *copy* a model [11], which is a different concern from inferring its training data.

> *Research question: can an adversary leverage model explanations to infer private information about the training data, and which kinds of explanations make this possible?*

The paper answers this by attacking three families of explanations. For backpropagation-based explanations it measures membership leakage through a simple threshold attack and investigates *why* the leakage arises. For perturbation-based explanations it shows that the attack fails and relates this to explanation quality. For example-based explanations (in the appendix) it demonstrates membership inference with certainty and full dataset reconstruction. The authors describe this as the first extensive analysis of the privacy risks of explanations, and they argue that transparency and privacy may have to be traded against each other.

2 Background

2.1 Model explanations

A classifier c maps a point $x \in \mathbb{R}^n$ to a distribution over k labels. A *feature-based* explanation $\varphi(y)$ assigns to each feature i of a point of interest y a value $\varphi_i(y)$ expressing its influence on the prediction. The paper studies two groups:

- **Backpropagation-based** explanations use a few backward passes. The canonical one is the *gradient* $\varphi_{\text{GRAD}}(y) = \partial c/\partial x\,(y)$ [the "saliency map"]. *Integrated Gradients* [9] averages gradients along a path from a baseline to y and satisfies completeness (attributions sum to $c(y) - c(\text{baseline})$); *Guided Backpropagation*

keeps only positive evidence through ReLU units; *Layer-wise Relevance Propagation* (LRP) redistributes the output layer by layer back to the inputs.

- **Perturbation-based** explanations query the model on many modified inputs. *SmoothGrad* [7] averages the gradient over copies of y with Gaussian noise; *LIME* [8] fits a sparse local linear surrogate model on samples around y.

An **example-based** explanation instead returns training points. The influence-function method of Koh and Liang [6] returns the k training points x with the largest influence $I_y(x) = L(y, \theta_{-x}) - L(y, \theta)$, i.e. the change in the loss at y when x is removed from training.

2.2 Membership inference and threat model

The adversary holds a set of candidate points, half of which were used for training, so random guessing achieves 50% accuracy. Models usually have a lower loss on training points; the classic *loss-threshold attack* [3, 4] predicts "member" if $L(x, \theta) \le \tau$ and is Bayes-optimal under some assumptions [4]. It requires the true label, which an adversary who only observes explanations may not have. In the paper's threat model the adversary submits a query x and receives the prediction and the explanation $\varphi(x)$. Following Kerckhoffs's principle, the adversary knows the architecture and training hyperparameters, and in the realistic setting it has some data from the same distribution.

3 Methodology

3.1 Threshold attacks on explanation variance

The central proposal generalises the loss-threshold attack to other signals via the variance of a vector, $\mathrm{Var}(v) = \Sigma_i (v_i - \mu_v)^2$. Two new attacks are defined:

$$Membership_{Pred}(x) = [Var(c(x)) \ge \tau], \quad Membership_{Expl}(x) = [Var(\varphi(x)) \le \tau].$$

The intuition is about distance to the decision boundary. Loss minimisation pushes the decision boundary away from training points, so around a member the model output is flat and saturated: a small change to any feature barely changes the prediction, the gradient is small, and the explanation has low variance. Points close to the boundary, where tanh, sigmoid and softmax outputs are steep, are more often non-members and produce large, high-variance gradients. A confident prediction vector, conversely, has high variance, which explains the direction of the prediction attack. A high explanation variance thus acts as a label-free proxy for a high loss. The authors note that the ℓ_1 norm of the explanation worked equally well.

3.2 Choosing the threshold

Two calibrations are compared. The *optimal threshold* τ_{opt} is computed with knowledge of true membership; it is not a feasible attack but an estimate of worst-case leakage. The *shadow-model* attack trains $s \in \{1, 3\}$ models on disjoint data from the same distribution and uses the threshold that is optimal for them, which is the practical attack of Shokri et al. [2].

3.3 Learned attack on the full explanation

To test whether the variance discards useful information, a neural attack model (layers [r, 1024, 512, 64, 256, 64, 1], trained on 20,000 explanations of known members and non-members) receives the entire explanation vector. The same framework is used to combine explanation, prediction and loss as inputs.

3.4 Experimental setup

Six datasets are used (Table 1). Each is merged, split into four disjoint subsets, and each subset is split 50/50 into training and test data; four target models are trained, each can serve as shadow model for the others, and the procedure is repeated 25 times (100 attacks per dataset). Architectures follow earlier membership-inference work: fully connected tanh networks for the tabular data, a small CNN for CIFAR-10 and an AlexNet variant for CIFAR-100. Explanations are computed with the iNNvestigate library.

Dataset	Records	Features	Classes	Train acc.	Test acc.
Purchase (shopping histories)	197,324	600	100	1.00	0.75
Texas (hospital stays)	67,330	6,170	100	0.98	0.52
CIFAR-100	60,000	3,072	100	0.97	0.29
CIFAR-10	60,000	3,072	10	0.93	0.53
Diabetic Hospital	101,766	127	2	0.64	0.61
UCI Adult (census)	48,842	104*	2	0.85	0.85

Table 1. Target datasets and average accuracies of the target models (from Tables 1–2 of [1]). *Adult has 14 raw attributes, one-hot encoded to 104 features. The gap between training and test accuracy shows how strongly most targets overfit.

3.5 Analysis experiments

To explain *when* leakage occurs, the authors generate synthetic classification data with scikit-learn's `make_classification`, varying the number of features from 1 to 10,000 with 2 or 100 classes, and measure the correlation between $\|\nabla c(x)\|_1$ and membership. They also vary the number of training epochs to control overfitting, and run the attack against SmoothGrad and, in a smaller setting (1,000 training points, 20 repetitions) because of its cost, against LIME.

3.6 Attacks on example-based explanations

For influence-function explanations (logistic regression on Adult and Hospital, and on Inception-v3 features for a Dog/Fish image dataset), membership inference is direct: a training point that appears among the top-k explanations of its own prediction is a certain member ("k-self-revealing"). The authors then target **dataset reconstruction**. Their Algorithm 1 exploits the fact that for logistic regression the influence of a training point is zero on an affine subspace: after a point is revealed, the attacker restricts further queries to the subspace where all previously revealed points have zero influence, forcing a new point to appear. They prove that the algorithm reveals at least d points, where d is the largest number for which the leave-one-out weight vectors are d-wise linearly independent, and that it is optimal in the worst case. A second, heuristic attack queries previously revealed points; its success depends on the *influence graph* in which each training point links to its k explanation points.

4 Results

4.1 Backpropagation-based explanations leak membership on some datasets

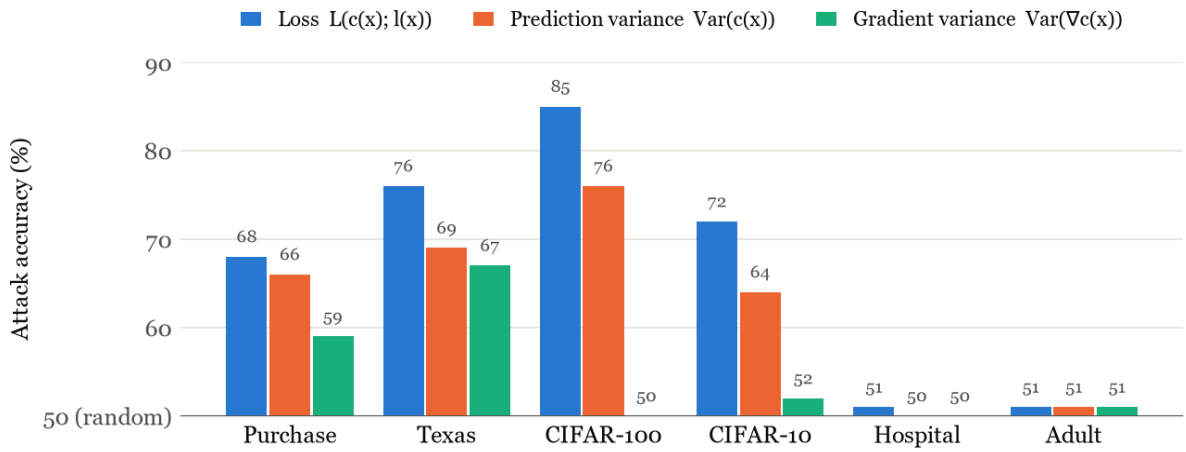


Figure 1. Accuracy of optimal-threshold membership attacks using the loss, the prediction variance or the gradient-explanation variance (re-plotted from Fig. 1 of [1]; values read approximately from the published chart). Bars start at the 50% random-guess level. Gradient explanations leak on Purchase and Texas, never more than the loss.

Figure 1 summarises the main experiment. Gradient explanations alone yield 59% accuracy on Purchase and 67% on Texas, which the authors call "a clear proof of concept" that explanations can be exploited. On the

image datasets the gradient attack is close to random, because gradient variance fluctuates strongly between individual images. On the low-dimensional Adult and Hospital data none of the attacks, not even the loss attack, beats random guessing; these targets hardly overfit. Attacks with shadow models come close to the optimal threshold, and with three shadow models they are nearly optimal, so the leakage is achievable by a realistic adversary. The prediction-variance attack is almost as strong as the loss attack, which shows that knowing the true label is not essential.

Giving a learned attack the entire gradient vector improves accuracy (Purchase from about 59% to 67%, Texas from 67% to 70%) but stays close to 50% on CIFAR-10, Adult and Hospital. The authors describe the gain as slight, but eight percentage points on Purchase is not small, so the variance does throw away part of the signal. Combining explanation, prediction and loss gives the same accuracy as the strongest single source; when the loss is available, explanations add nothing. Integrated Gradients, LRP and Guided Backpropagation behave like the plain gradient, with Integrated Gradients closest.

4.2 Dimensionality and overfitting determine leakage

On synthetic data the correlation between gradient norm and membership is essentially zero up to about 100 features, becomes clearly negative (members have smaller gradients) between roughly 10^3 and 10^4 features, and weakens again at the highest dimensions where training accuracy drops and the model fails to learn. Dense, low-dimensional data give members and non-members similar distances to the boundary, which matches the failure on Adult and Hospital; the number of classes plays a minor role, and smaller or larger networks shift the interval without changing the pattern. Overfitting increases leakage: on Texas the gradient attack rises from about 55% to 67% over 50 epochs and on Purchase from about 53% to 59%, while CIFAR-10 stays near 52%.

4.3 Perturbation-based explanations do not leak, at a cost

Neither threshold nor learned attacks beat random guessing when given SmoothGrad or LIME explanations. The authors conjecture that this is because these methods evaluate the model on points off the data manifold, where its behaviour is unspecified and carries no information about the training set. The same property has been criticised because it lowers the fidelity of the explanation and makes it manipulable [10]; the authors also point out that making models robust to such perturbations is known to increase membership leakage [18]. They argue that an explanation of a model that effectively memorises its labels *should* reflect that memorisation, so the privacy of these methods may be a symptom of low explanation quality. In the small-scale rerun, the gradient attack unexpectedly reached about 84% on Texas but only about 51% on Purchase, an anomaly the authors cannot explain.

4.4 Example-based explanations leak membership and whole datasets

For influence-based explanations, releasing the top 5 points reveals on average about a quarter of each training set as self-revealing, and 63% for Hospital; even releasing only the single most influential point reveals 10% of Adult members. The subspace-reduction attack recovers 1,790 of 1,800 Dog/Fish training images (99.4%), but only 92 of 2,000 Adult records (4.6%) and 81 of 2,000 Hospital records (4.0%). The algorithm cannot recover more points than the data dimension (n = 104 and 127), so on Adult it gets close to its limit. The influence graphs are fragmented (for k = 5, between 1,448 and 1,742 strongly connected components per training set, most of size 1, and up to 1,568 points with in-degree 0), so the adaptive heuristic recovers roughly the largest component (up to 228 points) and needs about twice as many queries as an omniscient greedy attacker.

Group	Share of data	$k = 1$	$k = 5$	$k = 10$
Dog/Fish: whole dataset	100%	16%	24%	28%
Dog/Fish: birds	0.6%	64%	84%	86%
Hospital: whole dataset	100%	33%	63%	75%
Hospital: age 0–10	0.2%	61%	100%	100%
Hospital: Asian American	0.6%	33%	70%	95%

Table 2. Share of training points revealed as their own explanation when the top-k influential points are released (selection from Table 3 of [1]). Small groups and outliers are revealed far more often than average.

The result I found most worrying is in Table 2. Minorities and outliers are exposed far more often than average, because unusual points are more influential and more likely to be memorised. The Adult dataset is a partial exception: young people and Black individuals are revealed less often, which the authors attribute to these attributes being highly predictive of the label there.

5 Critical analysis

5.1 Strengths

The biggest contribution of the paper is that it started this line of research. Before it, explanations were discussed almost only as a transparency tool, and the paper made the privacy cost concrete for both the machine-learning and the policy community. The evaluation is broad: six real datasets, seven explanation methods, two ways of setting the threshold, a learned attack, and a separate theoretical part on example-based explanations with proven guarantees.

What I liked most is that the authors do not stop at reporting attack accuracy. They try to explain *when* leakage happens (distance to the decision boundary, data dimensionality, overfitting) and test this with controlled synthetic experiments, and the minority analysis connects privacy to fairness. The attack itself is also very simple. It needs no true labels and works almost as well with shadow models as with the optimal threshold, so it is a realistic threat and not only a worst-case bound.

5.2 Limitations

My main concern is the evaluation metric. All results are average accuracy on balanced sets of members and non-members. Carlini et al. [5] later showed that average accuracy can hide whether an attack identifies *any* individual with confidence, and recommended reporting the true-positive rate at a low false-positive rate instead. A 67% accuracy could come from a small set of records that are identified with near certainty, or from a weak signal spread over everyone. The paper cannot tell these apart, and for judging the real privacy harm the difference matters.

Second, for feature-based explanations the explanation never beats the loss or prediction attack, and combining sources adds nothing, so the explanation mostly acts as a stand-in for the loss. The added risk only really exists for a service that releases labels and explanations but hides confidence scores. I think the paper could have said this more directly. Third, most target models overfit heavily (Texas: 98% training versus 52% test accuracy; CIFAR-100: 97% versus 29%). A well-regularised production model would probably leak less, and no regularised or differentially private target is evaluated.

The perturbation-based results are the thinnest part. The off-manifold explanation for why SmoothGrad and LIME resist the attack is a conjecture, and it could have been tested, for example by varying SmoothGrad's noise level σ. The LIME experiment is small, and the unexplained 84% gradient result on Texas in that setting suggests that results vary a lot between setups; no error bars are reported anywhere in the paper. The example-based results cover only logistic regression and last-layer models with about 2,000 training points and exact leave-one-out retraining, not the approximations used in practice, and the minority analysis rests on groups of 0.2–0.6% of the data, i.e. a handful of records. Finally, differential privacy and adversarial regularisation [12, 19] are mentioned but not evaluated, and the explanation-quality side of the proposed trade-off is never measured.

6 Discussion

Here I go through the points I bring up in the seminar presentation and how the paper connects to the rest of the course.

6.1 Explanations as a release about private data

In the terms used in the course, an explanation is one more answer released about a statistical model of personal data, and membership inference is the standard way to audit such a release. The results fit the general lesson that every released statistic costs some privacy, and that the risk is highest for outliers, which are also the individuals that are hardest to protect with k-anonymity or aggregation.

6.2 What differential privacy would and would not fix

A gradient explanation $\partial c/\partial x(y)$ depends only on the trained model and the query y. If the model is trained with differential privacy, e.g. with DP-SGD [12], the post-processing property of DP means that gradient, Integrated Gradients, LRP, SmoothGrad and LIME explanations get the same guarantee at no extra privacy cost. Example-based explanations are different: they access the training data directly when the explanation is computed, so post-processing does not apply and they need their own mechanism, which is what later work on differentially private explanations develops [13]. In my view this is also the simplest way to see why example-based explanations did so much worse in the paper.

6.3 Connection to Bertran et al.

Bertran et al. [16] show that when a user exercises the right to erasure, the difference between the models before and after unlearning, multiplied by an estimated Hessian, is approximately the gradient of the deleted record, and the record can be reconstructed from it even for linear regression. This is the same influence-function mathematics that Koh and Liang's explanations [6] are built on. The two papers make a similar point: a mechanism added to serve a data-protection right (explanation under GDPR Articles 13–15 and 22, erasure under Article 17) releases a second view of the model, and that second view leaks information that the model alone did not. Both point to differential privacy as the defence, but neither paper evaluates it. There is also a parallel in the results. Bertran et al. find that membership inference against their linear models is at random-guess level, just like all attacks on Adult and Hospital here, and yet in both papers the extra release makes reconstruction possible for exactly these simple models.

6.4 Questions from the seminar

Pablo Louis Henri Pierre, who was assigned to ask questions about this paper, asked two. *How can an attacker use model explanations to steal private information?* For feature-based explanations, the attacker queries a record it suspects was a training point and predicts "member" if the explanation's variance is below a threshold calibrated on shadow models (Section 3.1). Training points lie far from the decision boundary, where explanations are small and flat, so no true label is needed. Example-based explanations return training records themselves: a record that explains its own prediction is a certain member, and querying where already revealed points have no influence recovered 99.4% of the Dog/Fish training images (Section 4.4).

Are some types of model explanations more dangerous than others? Yes. Example-based explanations are the most dangerous: they release actual records, give certainty instead of a statistical guess and expose minorities first (Table 2). Backpropagation-based explanations leak moderately, mainly for high-dimensional, overfitted models, and never more than the prediction vector or the loss. SmoothGrad and LIME did not leak, probably because they are less faithful to the model (Section 4.3). Differential privacy gives the same ranking (Section 6.2).

I had also prepared for three other questions. The first is whether 59–67% accuracy is a serious risk. As an average over balanced sets it is moderate; what matters is whether specific individuals can be identified with few false positives, and the paper does not measure that. The second is whether a service could simply hide confidence scores. That does not help much, because the gradient explanation carries almost the same signal as the prediction vector. The third is whether services should stop offering explanations. The authors reject this. They argue for explanation methods with a measured privacy–quality trade-off, since releasing random noise is private but useless, while example-based explanations are useful but unsafe.

7 Future work and conclusion

The most direct follow-up is explanation methods with formal differential-privacy guarantees, together with a measurement of how much fidelity they lose; Patel et al. [13] have started this for feature-based and counterfactual explanations. The attacks should also be re-evaluated with per-example calibration and TPR at low FPR [5], on regularised and DP-trained models, and on modern architectures such as transformers and large language models. Other explanation types leak as well: counterfactual explanations and algorithmic recourse reveal membership [14], and Liu et al. [15] study how the amount of detail in an explanation affects membership inference, so a common framework covering all explanation types would help practitioners.

Two further directions follow from the discussion above. Explanations and unlearning are often offered by the same service, and the joint leakage when an adversary sees both explanations and deletion updates, in the spirit of [16], has not been studied as far as I know. The disparate exposure of minorities also deserves a proper study with larger groups, since transparency is partly meant to protect these groups in the first place.

To conclude, Shokri, Strobel and Zick show that backpropagation-based explanations leak membership roughly as much as the model's own confidence scores, that perturbation-based explanations seem private mainly because they are less faithful to the model, and that example-based explanations can reveal most of a training set, with minorities exposed first. The quantitative evidence for feature-based explanations is weaker than the headline numbers suggest, but later work [14, 15] has found similar leakage. My main takeaway is that explanations should be counted as part of what a model reveals about its training data, in the same way as predictions and confidence scores.

References

1. R. Shokri, M. Strobel, Y. Zick. On the privacy risks of model explanations. *Proc. AAAI/ACM Conf. on AI, Ethics, and Society (AIES)*, 2021.
2. R. Shokri, M. Stronati, C. Song, V. Shmatikov. Membership inference attacks against machine learning models. *IEEE S&P*, 2017.
3. S. Yeom, I. Giacomelli, M. Fredrikson, S. Jha. Privacy risk in machine learning: analyzing the connection to overfitting. *IEEE CSF*, 2018.
4. A. Sablayrolles, M. Douze, Y. Ollivier, C. Schmid, H. Jégou. White-box vs black-box: Bayes optimal strategies for membership inference. *ICML*, 2019.
5. N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, F. Tramèr. Membership inference attacks from first principles. *IEEE S&P*, 2022.
6. P. W. Koh, P. Liang. Understanding black-box predictions via influence functions. *ICML*, 2017.
7. D. Smilkov, N. Thorat, B. Kim, F. Viégas, M. Wattenberg. SmoothGrad: removing noise by adding noise. *arXiv:1706.03825*, 2017.
8. M. T. Ribeiro, S. Singh, C. Guestrin. "Why should I trust you?": explaining the predictions of any classifier. *KDD*, 2016.
9. M. Sundararajan, A. Taly, Q. Yan. Axiomatic attribution for deep networks. *ICML*, 2017.
10. D. Slack, S. Hilgard, E. Jia, S. Singh, H. Lakkaraju. Fooling LIME and SHAP: adversarial attacks on post hoc explanation methods. *AIES*, 2020.
11. S. Milli, L. Schmidt, A. D. Dragan, M. Hardt. Model reconstruction from model explanations. *ACM FAT**, 2019.
12. M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, L. Zhang. Deep learning with differential privacy. *ACM CCS*, 2016.
13. N. Patel, R. Shokri, Y. Zick. Model explanations with differential privacy. *ACM FAccT*, 2022.
14. M. Pawelczyk, H. Lakkaraju, S. Neel. On the privacy risks of algorithmic recourse. *AISTATS*, 2023.
15. H. Liu, Y. Wu, Z. Yu, N. Zhang. Please tell me more: privacy impact of explainability through the lens of membership inference attack. *IEEE S&P*, 2024.
16. M. Bertran, S. Tang, M. Kearns, J. Morgenstern, A. Roth, Z. S. Wu. Reconstruction attacks on machine unlearning: simple models are vulnerable. *NeurIPS*, 2024.
17. B. Goodman, S. Flaxman. European Union regulations on algorithmic decision-making and a "right to explanation". *AI Magazine* 38(3), 2017.
18. L. Song, R. Shokri, P. Mittal. Privacy risks of securing machine learning models against adversarial examples. *ACM CCS*, 2019.
19. M. Nasr, R. Shokri, A. Houmansadr. Machine learning with membership privacy using adversarial regularization. *ACM CCS*, 2018.